

Effects of Forced Induction Systems on Internal Combustion Engines

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Abstract

Forced induction systems can increase the power output from internal combustion engines by increasing the oxygen content per unit volume that is aspirated by the engine. This is beneficial to automotive manufacturers because it allows them to fit smaller engines with forced induction systems into vehicles and produce comparable power outputs to naturally aspirated engines with a larger displacement. Forced induction systems also decrease the emissions produced from engines which helps automotive manufacturers keep their vehicles within emissions standards while being able to provide the power outputs that their customers want. While forced induction systems do benefit manufacturers for the reasons stated previously, they increase the cost of engine development and manufacturing and add to maintenance. These systems contain parts that move at extreme speeds and failure leads to a complete engine failure. That being said, it needs to be assessed per use case whether a forced induction system makes sense and which type of system is best suited for that use case.

An internal combustion engine runs on small, perfectly timed, explosions that result in a piston being forced down to drive a crankshaft. A naturally aspirated engine provides air at atmospheric pressure to the intake manifold where it is then mixed with fuel further in the engine through the opening of valves to allow the air into the combustion chamber[3]. The fuel injection is timed in order to create an air fuel ratio that is optimized for the engine. The problem with natural aspiration is that engines need to be built to larger displacements in order to output comparable power to a forced induction engine.

Forced induction is the process of providing air at higher than atmospheric pressure to the engine. More fuel can be injected as a result and this creates larger combustions. The two proven means of forced induction are turbocharging and supercharging, which each have their own advantages and disadvantages. The main drive for forced induction systems was the development of high power engines, but many manufacturers are using these systems today to provide smaller engine packages and meet emissions standards[9]. Goals to further compact engine packages and reduce emissions has driven the development of electrical forced induction systems as well.

Turbochargers increase air pressure and volume into the engine by utilizing the pressure of exhaust gasses from the combustion chamber of the engine to spin a turbine. This turbine is connected via a shaft

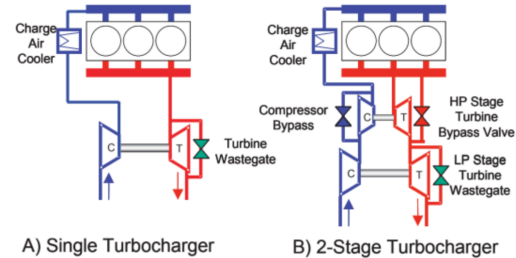


Figure 1: Forced Induction Systems Schematics[11]

to another turbine that pulls air into the intake. Figure 1 shows two layouts for forced induction systems of various components. System A shows a single turbocharger being used; the exhaust gasses turn the turbines which pull air in through the intake and then into an intercooler. Coolant is run through the intercooler and exchanges heat with the air being pushed through, causing the air to decrease in volume and increase in oxygen content per unit volume. In most engines the ideal air:fuel ratio is around 14.7:1 by weight, meaning if more air can be inducted into the engine via pressure and cooling, more fuel must be added as well[2]. The increase in oxygen content via cooling and higher intake pressure of the intake air allows more fuel to be injected and a larger combustion reaction to take place. For the pressure on the intake side of the turbo to increase, the pressure

must increase on the exhaust side as well. The main issue with single turbo systems is the amount of time that it takes to increase the boost pressure and turn it into power. The engine must create sufficient exhaust pressure to increase the turbine speed in the compressor to its maximum efficiency range to create boost for the engine[5]. The engine needs to be accelerated to high rpm's before sufficient exhaust pressure is created to accelerate the compressor's turbine to this range. Uthoff and Yakimow, of Eaton Corp., have found that turbochargers using variable geometry housing and ceramic turbines still take up to four times as long to provide maximum boost pressure as a positive displacement supercharger[8]. Nonetheless, turbochargers have been continuously advanced in order to provide more power at lower speeds and decrease engine package size.

System B in Figure 1 shows a twin-turbo system that aims to increase power at lower engine speeds by using two turbochargers of different sizes. This setup creates increased intake pressures from the smaller turbocharger at lower exhaust pressures and even higher intake pressures from the larger turbocharger at higher exhaust pressures[11]. A single turbocharger system must use a turbine that is too large to create high boost at low exhaust pressures and too small to create high boost at high exhaust pressures. It must be sized in the mean that gives enough boost in the low end and high end to be useable but does not provide exceptionally high boost in both. To combat this issue the twin-turbo system uses turbochargers of different sizes. One turbocharger has a small turbine that can be accelerated to a higher angular velocity at lower exhaust pressures. This turbocharger, in turn, has a compressor with a smaller turbine that cannot produce high enough intake pressure for the air capacity needed at higher engine speeds. The second turbocharger has a larger turbine that provides adequate intake pressure at higher engine speeds and higher exhaust pressures. The twin-turbo system has allowed vehicle manufacturers to produce vehicles with engines of smaller displacement which decreases total fuel consumption and costs[6]. By decreasing the total fuel consumption of the engine, emissions are decreased all while keeping consumer vehicles powerful enough for daily driving. Figure 2 shows the difference of efficiency between single and double-stage turbocharging system at low exhaust pressures. This model clearly shows the increased efficiency of a dual-stage turbocharger system at low pressures while Figure 3 shows the two systems at high pressures. The high pressure model is only measuring the larger turbocharger in the dual-stage system but shows the

larger turbine allows for boost to be maintained at higher exhaust pressures than the single-stage turbocharger system. These models show that the dual-stage turbocharger system helps reduce some of the time before intake pressure is increased by the forced induction system thus reducing the 'lag' of power exhibited by single-stage turbocharger systems.

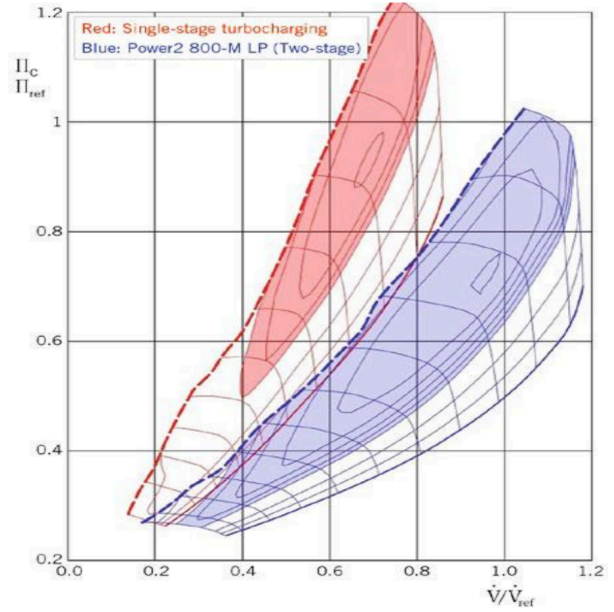


Figure 2: Efficiency of Single-Stage and Double Stage Turbocharging Systems at Low Pressure(Exhaust Output vs. Intake Induction)[6]

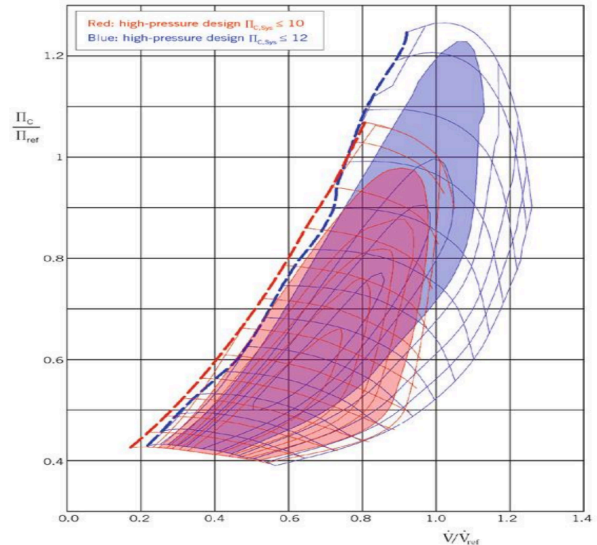


Figure 3: Efficiency of Single-Stage and Double Stage Turbocharging Systems at High Pressure(Exhaust Output vs. Intake Induction)[6]

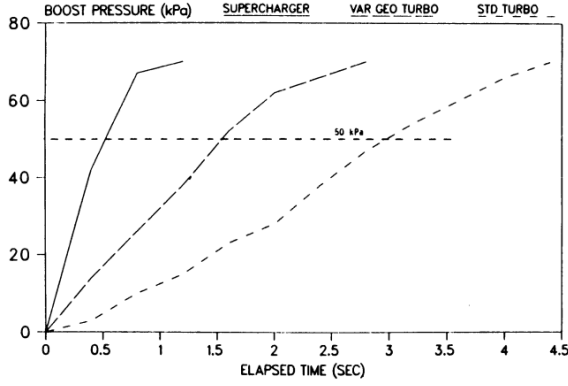


Figure 4: Response Time of Various Forced Induction Systems[8]

Superchargers aim to reduce the time for intake pressure to increase by getting their power to turn the turbine from the engine's crankshaft instead of exhaust pressure from the engine. With the turbine shaft being connected directly to the crankshaft via a belt or chain, this causes the turbine to rotate at a speed proportional to the engine speed at all times. Because the turbine is always rotating at a speed proportional to the engine speed, the intake pressure is high enough at all engine speeds to provide boost (even at low engine speeds which is an improvement over turbochargers). Figure 4 shows the difference in response time between an Eaton Model 45 supercharger, a standard turbocharger based on the engine specs, and a variable geometry turbocharger with a ceramic turbine wheel. The model shows the drastic difference in boost pressure between a supercharger and turbocharger for any given time after the throttle is applied[8]. The supercharger takes only 0.4 seconds to produce 50 kPa of intake pressure while the variable geometry turbocharger takes closer to 1.5 seconds. The immediate boost pressure that a supercharger makes available increases the throttle response of the vehicle and increases power at all engine speeds. Superchargers are typically internally liquid-cooled in order to cool the intake air and increase the oxygen content just as the intercooler of a turbocharger system does. Superchargers, mounted on the intake manifold, have a thermal advantage because they cool the air going into the intake right before it enters as opposed to the air running through hot pipes after the intercooler and into the engine as a turbocharger system is setup. This makes the compressor efficient in cooling the air and compressing it without adding much heat back in through the compression process[1]. Figure 5 shows the adiabatic ef-

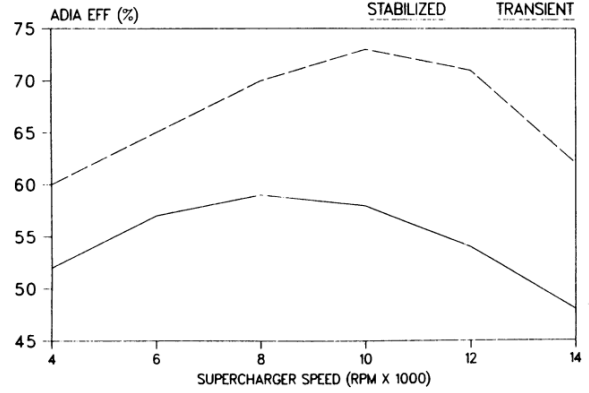


Figure 5: Adiabatic Efficiency of Eaton Model 45 Supercharger[8]

iciency of an Eaton Model 45 supercharger at different turbine speeds. The transient state better models what would be expected during normal vehicle use while the stabilized model shows the efficiency at prolonged periods of individual turbine speeds. In a transient state, this model shows between 60% and 73% adiabatic efficiency for this model supercharger which means the air is compressed with up to a 73% efficiency in lack of heat exchange. A higher efficiency is wanted in order to reduce the intrinsic heating of the air when it is compressed which would decrease the oxygen content per unit volume being inducted into the engine. Another advantage to a supercharger is the length of the shaft connecting the pulley to the turbine is much shorter than the shaft connecting the exhaust and compressor turbines of a turbocharger[7]. Vibrations are intrinsic to any body rotating on a bearing but the moments created by these forces are decreased if the moment arm is decreased. The bearings used in both superchargers and turbochargers are typically floating ring bearings which demonstrate highly non-linear behavior due to their construction. However, with superchargers having generally smaller turbines than turbochargers and shorter shafts between their loads, the forces on the bearing are less in the supercharger which creates a product with less maintenance and higher quality. Superchargers, just like turbochargers allow vehicle manufacturers to decrease the engine package size and create environmentally and economically efficient vehicles.

Forced induction is not only beneficial to creating more power. Forced induction systems have been studied extensively in order to decrease emissions of carbon dioxide and nitrogen oxides. Equation (1) shows a non balanced chemical equation for the com-

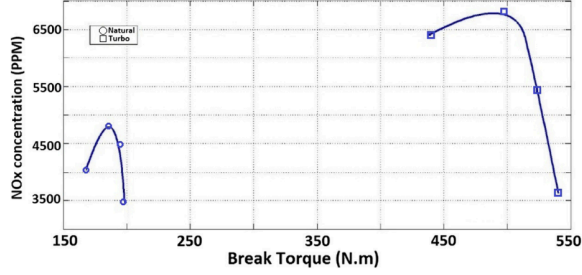


Figure 6: Nitrogen Oxides' Concentration vs. Brake Torque[4]

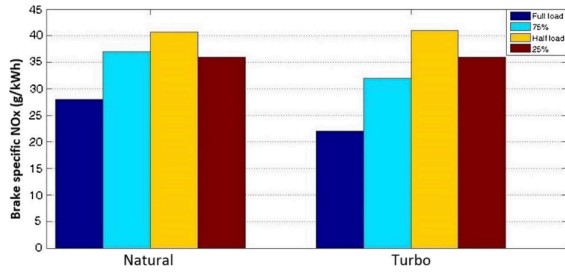
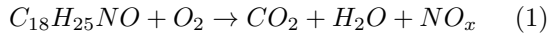


Figure 7: Normalized Nitrogen Oxides' Concentration in an Engine at 4500rpm[4]

bustion of indolene, a fuel commonly of highest abundance in pump gases[4].



The recent goal for vehicle manufacturers has been to reduce the amount of carbon dioxide and nitrogen oxides that are emitted into the atmosphere. Figure 6 shows the difference of a naturally aspirated and turbocharged engine and the nitrogen oxides produced at different brake torques. This figure, at a quick glance may lead one to believe that the emissions of nitrogen oxides are much higher in a turbocharged engine but this is at a significantly higher brake torque. If the data is normalized to show brake specific emissions levels, as it is in Figure 7, the difference between a naturally aspirated engine and turbocharged engine becomes more balanced. The balanced model is derived by dividing the mass flow of the nitrogen oxides in g/h by the power in kW[4]. The emissions improvements from a turbocharged system shown in Figure 7 are due to more of the hydrocarbons and amines in the fuel being burned due to the increased oxygen content going into the intake manifold[10], [4]. In recent efforts to increase emissions efficiency and decrease fuel consumption even more, electric superchargers and turbochargers are being researched. In an electric turbocharger system, the compressor tur-

bine is connected directly to a high speed electric motor. The compressor turbine speed can in turn be controlled to be directly proportional to the engine speed through logic circuits between the ECM and turbocharger motor control unit, independent of the exhaust turbine[10]. Electronic turbochargers are being developed in two manners: one being a hybrid system and the other being a standalone electronic unit. The hybrid system utilizes exhaust pressure to turn the turbines, but gets help from the electronic motor at low engine speeds and exhaust pressures. The standalone system does not depend on the exhaust pressure at all and is solely driven by an electronic motor. The standalone system as seen up to 35% increases in transient response times from traditional turbocharger systems and the emissions efficiency is hypothesized to increase as well but is still being tested[10].

In summary, forced induction systems can increase the power output from internal combustion engines by increasing the oxygen content per unit volume that is aspirated by the engine. This is beneficial to automotive manufacturers because it allows them to fit smaller engines with forced induction systems into vehicles and produce comparable power outputs to naturally aspirated engines with a larger displacement. Forced induction systems also decrease the emissions produced from engines which helps automotive manufacturers keep their vehicles within emissions standards while being able to provide the power outputs that their customers want. While forced induction systems do benefit manufacturers for the reasons stated previously, they increase the cost of engine development and manufacturing and add to maintenance. These systems contain parts that move at extreme speeds and failure leads to a complete engine failure. That being said, it needs to be assessed per use case whether a forced induction system makes sense and which type of system is best suited for that use case.

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